

# Tomas Ulises Traini

## Software Engineer, Clinical Research Systems

Rosario, Argentina. Argentine and Spanish citizen, EU passport. Open to remote or relocation.  
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## SUMMARY

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Full-stack engineer, about five years, mostly .NET and React, now Python and FastAPI. At CLINIGMA in Denmark I am the only engineer on their clinical research tooling, which turns recorded patient interviews into analysis-ready qualitative data for clinical trials. Before that, React and SAP-backed microservices at John Deere, and .NET modernization and security work at PwC. English C2, German B1 and improving.

## SKILLS

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**Languages:** Python, TypeScript, C#, JavaScript, SQL

**Backend:** FastAPI, SQLAlchemy, Pydantic, .NET and .NET Core, NestJS, Node.js, REST, GraphQL, WebSockets

**Frontend:** React, TypeScript, Zustand, TanStack Query, Redux, Vite, Tailwind, Angular

**Data and cloud:** PostgreSQL, SQL Server, DynamoDB, Azure (App Service, Blob Storage, OpenAI), AWS SQS, Terraform, Docker, Azure DevOps

**Practice:** hexagonal architecture, microservices, pytest, vitest, Cypress, Playwright, AI agent workflows, WCAG 2.1, Checkmarx

**Clinical domain:** ICH-GCP, GxP, 21 CFR Part 11, audit trails, e-signature workflows, qualitative coding, codebooks, QDPX (REFI-QDA), speech recognition, speaker diarization

## EXPERIENCE

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### Technical Officer, Software Engineer (Consultant)

CLINIGMA ApS. Denmark, remote. Oct 2025 - Present

*CLINIGMA runs in-trial patient interviews and produces qualitative patient-experience data for pharmaceutical research and regulatory submissions, under ICH-GCP and ISO 9001.*

- Sole engineer on the company's clinical research tooling, which turns recorded patient interviews into analysis-ready qualitative data. I chose the architecture, the stack and the coding standards, with the CTO reviewing. Python and FastAPI on the backend, React and TypeScript on the client, PostgreSQL, running on Azure. In internal pilot since September 2026.
- The binding constraint in this domain is provenance rather than throughput: who changed what, when, on whose authority, and whether you can still show it a year later. Multi-stage review with sign-off between stages, and a versioning model that keeps every revision recoverable and attributable.
- Speech-to-text pipeline work: word-level timestamps, speaker diarization, and keeping audio aligned with the text after people have edited it. Built a benchmarking harness to compare ASR backends on the same audio for word error rate and timing drift.
- Set the testing and CI discipline the codebase runs on. Several thousand tests across backend and front end plus end-to-end runs, enforced layer boundaries, file-size limits, locale parity, and a rule that every bug lands in a ledger with the test that catches it again.
- Built an internal document system on a reusable schema, then argued for killing it. The engineering held up. It asked users to decide whether a form was reusable or single use, and most of them work in Word, so it was the wrong thing to build.
- Ran agent orchestration experiments with a colleague: idea refinement with a human in the loop, an architect step, one implementation agent per stack, QA, then a pull request. Two users. The finding was that coordination cost decides whether any of it is worth doing.

*Further detail on this work is withheld deliberately. Happy to go deeper in an interview, within what my agreement allows.*

### Senior Software Developer

Sistran Consultores. Remote. Aug 2025 - Feb 2026

- React and .NET work on insurance policy workflows, inside an existing enterprise codebase and delivery process.

## Ssr. Software Engineer

John Deere. Remote. Sep 2023 - Jun 2025

- Main React and TypeScript engineer on the tractor parts management apps, in a Node.js microservices setup with Terraform and CI/CD. Worked through a large front-end refactor: reducer-based state instead of prop drilling, a proper type convention instead of any and over-generic types, consistent naming.
- Added GraphQL queries and pagination for catalogs where one SAP query could return around 100,000 rows. Rendering that naively killed the page, so performance set the design.
- Owned the features that crossed into SAP. Connected the parts service, coordinated the SAP developer so their events landed alongside ours, extended the events service on AWS SQS, and pushed live part-state changes to the front end over WebSockets. A part could be red in SAP and green in ours, so idempotency and race conditions were the real work. Wrote the flows up as use cases and handed them to QA.

## ReactJS Developer (Contractor)

Podium Pick'em Challenge. Remote. Apr 2024 - Jun 2024. Freelance, alongside John Deere.

- Responsive React front end with reusable components for a sports prediction platform tied to the 2024 Olympics, built to a tight deadline.

## Ssr. .NET Developer

PwC. Remote. Aug 2022 - Sep 2023

- Co-led a three-person team modernizing legacy .NET Framework systems for US clients. Visiting US leadership recognized the team for the drop in open vulnerabilities that year.
- I was the person people came to on application security. Cleared 30+ Checkmarx findings and worked out staged .NET and NuGet upgrade paths for codebases full of vulnerable packages, without rewriting them. One package problem went all the way to Microsoft's internal team.
- Features, maintenance and WCAG 2.1 accessibility on PwC Policy on Demand, which is still running. Later, Excel reporting for tax calculation work.

## .NET Developer

Ingenea S.R.L. Rosario, hybrid. Dec 2021 - Aug 2022

- .NET Core APIs and SQL Server stored procedures for clients in Argentina, Brazil and Mexico, Ternium among them, with data access split across repository and service layers.

## EDUCATION

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### Universidad Abierta Interamericana

Bachelor's Degree in Technology Information Management (Licenciatura en Gestión de Tecnología Informática). Apr 2025 to expected Nov 2027, in progress.

### Instituto Zona Oeste

Technical Degree in Software Development, a three-year technical program. Mar 2019 - Nov 2022. GPA 7.33.

## LANGUAGES

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Spanish, native. English, C2. German, B1 and working towards B2.